

Experiment 3-High Wind: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1922	13.287	14.664	21.805	2.1%	8.3%	14.0%	10.973	28.043	6.983	19.410	0.501	1.223	1.402
Proposed	1758	5.490	2.845	14.484	16.7%	38.9%	50.9%	25.078	74.399	12.251	33.399	0.757	1.566	1.678
PPO-GRU	1774	7.489	5.640	16.442	8.9%	25.6%	36.0%	24.614	66.570	13.017	32.159	0.758	1.513	1.674
MPC	442	7.053	2.745	22.489	7.2%	28.3%	54.1%	2.896	4.360	2.127	4.355	0.155	0.191	0.257

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1922	96.1%	78	3.9%	3.9%	77	98.7%	1	0
Proposed	2000	1758	87.9%	242	12.1%	11.7%	231	95.5%	11	0
PPO-GRU	2000	1774	88.7%	226	11.3%	10.3%	204	90.3%	22	0
MPC	2000	442	22.1%	1558	77.9%	77.9%	1558	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	1558	1487	95.4%	1321	88.8%	13.673
Proposed	1558	1320	84.7%	703	53.3%	5.929
PPO-GRU	1558	1340	86.0%	911	68.0%	7.992